Simulation Analysis

Sophie Litschwartz

2024-08-09

Introduction and Overview

*I need to write this once I figure out what goes here

Data Generating Process

The simulations are a fully factored experiment.

Block Level Fixed Covariates
Number of Blocks: 5; ATE: .2; ICC: .2; Average Number of Observations Per Cluster: 20; Proportion of Clusters Treated: .5 (this is fixed across blocks); Number of District Level Covariates: 0

Block Level Varying Covariates (Factors that are varied across simulations)

Three groups of block level parameters Blocks all the same, Blocks all messy, and Blocks all reasonable

Blocks all the same: How much do number clusters per block vary (size ratio)?: 0 (none); How much does the size of the cluster vary (size ratio)? 0 (none); ATE SD: 0; Correlation Between Treatment Effect and Cluster Size: 0; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: 0; Correlation of Proportion of Clusters Treated per Block and Impact: 0; Correlation Between Outcome Intercept and Cluster Size: 0

Blocks all messy: How much do number clusters per block vary (size ratio)?: 1.3 (big) ; How much does the size of the cluster vary (size ratio)? 1.3 (big); ATE SD: .6; Correlation Between Treatment Effect and Cluster Size: .8; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: .8; Correlation of Proportion of Clusters Treated per Block and Impact: .8; Correlation Between Outcome Intercept and Cluster Size: .8

Blocks all reasonable: How much do number clusters per block vary (size ratio)?: .4 (average); How much does the size of the cluster vary (size ratio)? .4 (average); ATE SD: .1; Correlation Between Treatment Effect and Cluster Size: .2; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: .1; Correlation of Proportion of Clusters Treated per Block and Impact: 0; Correlation Between Outcome Intercept and Cluster Size: .1

Each of the rest of these factors don't vary as a group but vary each individually and are fully factored with all other factors:

Average Number of Clusters per Block: 4,8,16

Cluster Level Varying Parameters

Correlation of Impact and Size of Cluster: 0, .8

Cluster Level Treatment Effect Standard Deviation: 0, .3

Cluster Level ICC: 0, .2

Number of Observations Per Cluster Varies: True, False

Ratio of Size Cluster Sizes: 0, .4, 1.3

Number of School-Level Covariates: 0, 1, 5

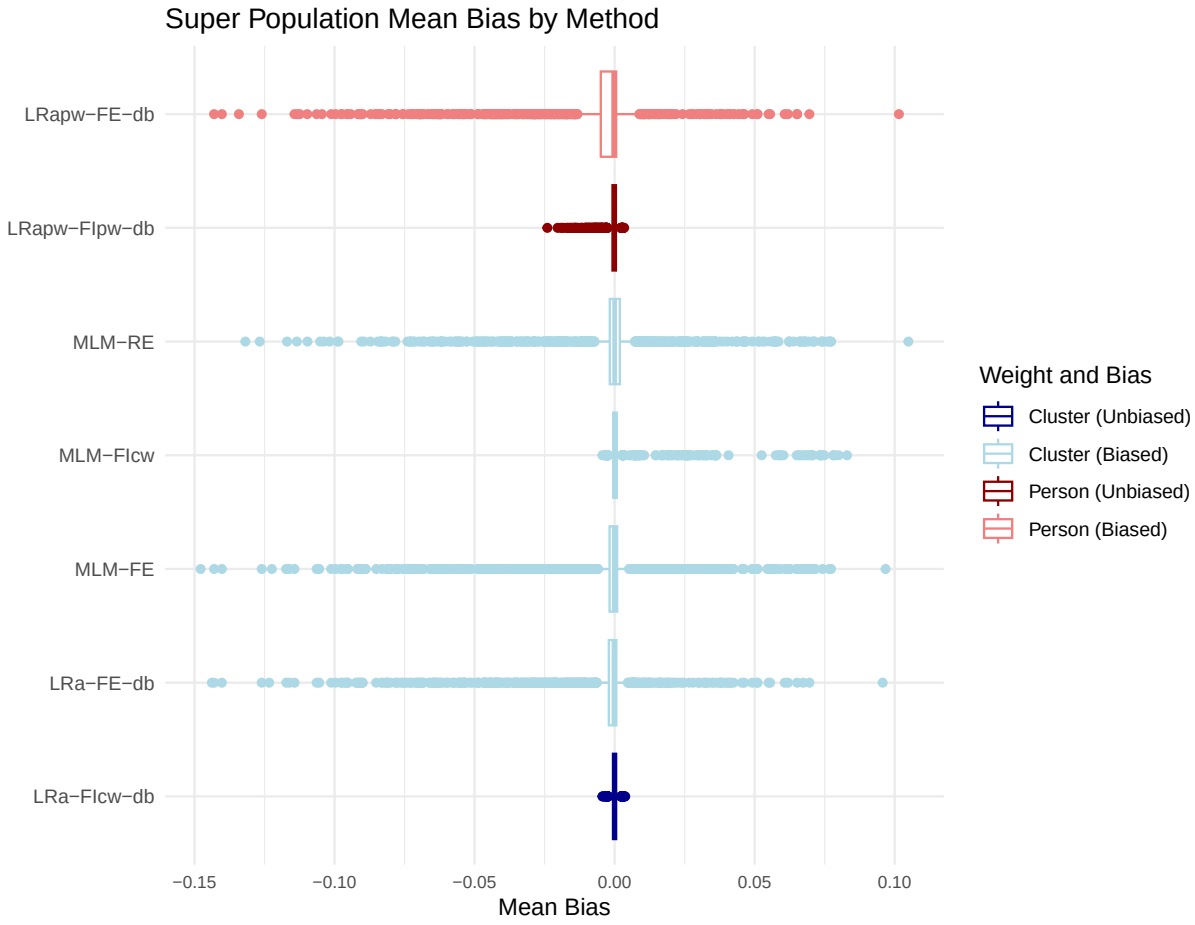
[School Level Covariate R Squared is always .5]

What does the bias look like for each estimator

The estimators that theory says should be biased are in fact much more biased across the simulations. These boxplots are across the different data generating processes, each dot is the mean bias across simulation replications. The methods in light blue are the ones theory says should be biased. I only show the finite population numbers because they are the same as the super population results.

There are tables showing how the different simulation facets predict bias. The regression model is how each facet of the data generating process predicts mean bias against a methods targeted estimand. The facets are binary indicators and are fully interacted with each other. I ran three models: just the cluster targeted methods, just the person targeted methods, and all methods. I don't know what to make of the results.

To-Do: Re-write the summary section



The Effect of Different Data Generating Parameters on the Magnitude of Mean Bias

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.007.	<.001*	.009*	.002***	.009*	.005**	.009*
block_typeS	<.001	<.001	<.001	<.001	<.001	<.001	.001
block_typeL	.036***	<.001	.037***	<.001	.042***	.001	.036***
J.barS	-.003	<.001	-.003	<.001	-.004	-.001	-.005
J.barL	-.002	<.001	-.002	.001**	-.004	-.002	-.003
size.impact.cor.schoolL	-.005	<.001	-.006*	-.003***	-.009**	-.008***	-.008**
sd.schoolL	<.001	<.001	-.002	-.002***	-.005	-.008***	-.007*
ICC.schoolL	-.007*	.001***	-.008*	.001**	<.001	.010***	<.001
variable.nTRUE	-.006.	<.001	-.007*	-.002***	-.010**	-.008***	-.009**
cv.schoolS	-.006	<.001	-.006.	-.001***	-.008.	-.004*	-.008*
cv.schoolL	-.001	<.001	-.002	-.002***	-.005	-.009***	-.007.
num.cov.schoolS	-.005	<.001	-.005	<.001	-.005	<.001	-.004
num.cov.schoolL	.001	<.001	.001	<.001	<.001	<.001	.001
block_typeS:J.barS	<.001	<.001	<.001	<.001	<.001	<.001	-.002
block_typeL:J.barS	<.001	<.001	<.001	<.001	-.001	<.001	.001
block_typeS:J.barL	<.001	<.001	<.001	<.001	<.001	<.001	-.003
block_typeL:J.barL	-.001	<.001	<.001	<.001	-.001	<.001	.004
block_typeS: size.impact.cor.schoolL	<.001	<.001.	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	-.006**	<.001	-.007**	<.001	-.011***	<.001	-.009***
block_typeS:sd.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:sd.schoolL	-.002	<.001	-.002	<.001	-.007**	<.001	-.007**
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	.002
block_typeL:ICC.schoolL	-.005*	<.001	-.004*	<.001	-.003	<.001	-.005*
block_typeS: variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: variable.nTRUE	.003	<.001	.001	<.001	-.002	<.001	-.002
block_typeS:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolS	.003	<.001	.002	<.001	.001	<.001	<.001
block_typeS:cv.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolL	.003	<.001	.001	<.001	-.003	<.001	-.004
block_typeS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolS	-.003	<.001	-.003	<.001	-.003	<.001	-.002
block_typeS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	.001
block_typeL: num.cov.schoolL	-.001	<.001	-.002	<.001	-.001	<.001	.001
J.barS: size.impact.cor.schoolL	.002	<.001	.002	<.001	.003	.001	.002
J.barL: size.impact.cor.schoolL	.003	<.001	.002	-.001***	.005.	.002	.004
J.barS:sd.schoolL	<.001	<.001	-.001	<.001	.001	.001	.002
J.barL:sd.schoolL	.001	<.001	<.001	-.001***	.003	.001	.003
J.barS:ICC.schoolL	.003	<.001	.003	<.001	.003	<.001	.002
J.barL:ICC.schoolL	.002	<.001*	.002	<.001	<.001	-.001	<.001
J.barS:variable.nTRUE	.003	<.001	.002	<.001	.004	.001	.003
J.barL:variable.nTRUE	.003	<.001	.002	-.001***	.005.	.001	.004.
J.barS:cv.schoolS	.002	<.001	.002	<.001	.002	.001	.003
J.barL:cv.schoolS	-.001	<.001	-.001	<.001	<.001	.001	.001
J.barS:cv.schoolL	-.002	<.001	-.002	<.001	-.001	.001	.001
J.barL:cv.schoolL	-.003	<.001.	-.004	-.001***	-.001	.001	<.001
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	-.001
J.barS:num.cov.schoolL	-.003	<.001	-.003	<.001	-.003	<.001	-.002
J.barL:num.cov.schoolL	-.003	<.001	-.002	<.001	-.004	<.001	-.005.
size.impact.cor.schoolL: sd.schoolL	.003	<.001	.005**	.003***	.010***	.011***	.010***
size.impact.cor.schoolL: ICC.schoolL	.002	<.001	.002	<.001	-.002	-.006***	-.002

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
size.impact.cor.schoolL: variable.nTRUE	<.001	<.001*	.002	.003***	.007***	.011***	.007***
size.impact.cor.schoolL: cv.schoolS	.002	<.001	.003	.001***	.005*	.005***	.005*
size.impact.cor.schoolL: cv.schoolL	.001	<.001	.003	.002***	.008**	.011***	.007**
size.impact.cor.schoolL: num.cov.schoolS	.001	<.001	.001	<.001	.001	<.001	.001
size.impact.cor.schoolL: num.cov.schoolL	-.001	<.001	-.001	<.001	<.001	<.001	-.001
sd.schoolL:ICC.schoolL	<.001	<.001	<.001	<.001	-.005*	-.006***	-.005**
sd.schoolL:variable.nTRUE	.001	<.001	.003	.003***	.008***	.011***	.008***
sd.schoolL:cv.schoolS	-.003	<.001	-.002	.001***	<.001	.005***	.002
sd.schoolL:cv.schoolL	-.002	<.001	<.001	.002***	.006*	.011***	.008***
sd.schoolL: num.cov.schoolS	.001	<.001	.001	<.001	.001	<.001	.002
sd.schoolL: num.cov.schoolL	-.002	<.001	-.002	<.001	-.001	<.001	<.001
ICC.schoolL: variable.nTRUE	.004*	<.001	.004*	<.001	-.001	-.006***	-.001
ICC.schoolL:cv.schoolS	.005*	<.001	.005*	<.001	.003	-.003**	.002
ICC.schoolL:cv.schoolL	.002	<.001	.002	<.001	-.003	-.006***	-.003
ICC.schoolL: num.cov.schoolS	.002	<.001	.003	<.001	.002	<.001	.002
ICC.schoolL: num.cov.schoolL	.001	<.001	.001	<.001	<.001	<.001	.001
variable.nTRUE:cv.schoolS	.001	<.001	.002	.001***	.004.	.004***	.005*
variable.nTRUE:cv.schoolL	.002	<.001.	.003	.003***	.009***	.011***	.010***
variable.nTRUE: num.cov.schoolS	<.001	<.001	.001	<.001	.001	<.001	<.001
variable.nTRUE: num.cov.schoolL	.002	<.001	.001	<.001	.002	<.001	<.001
cv.schoolS: num.cov.schoolS	.005.	<.001	.005.	<.001	.005.	<.001	.005.
cv.schoolL: num.cov.schoolS	.002	<.001	.002	<.001	.002	<.001	.002
cv.schoolS: num.cov.schoolL	.002	<.001	.002	<.001	.002	<.001	.001
cv.schoolL: num.cov.schoolL	.002	<.001	.001	<.001	.003	<.001	.002

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

met hod	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	-0.144	0	0	0	L	L	0	L
LRa-FE-db	-0.143	S	0	L	S	S	0	L
LRa-FE-db	-0.14	0	L	0	S	L	L	L
LRa-FE-db	-0.126	0	0	0	L	L	0	L
LRa-FE-db	-0.123	0	L	L	L	L	0	L
LRapw-FE-db	-0.143	S	0	L	S	S	0	L
LRapw-FE-db	-0.14	0	L	0	S	L	L	L

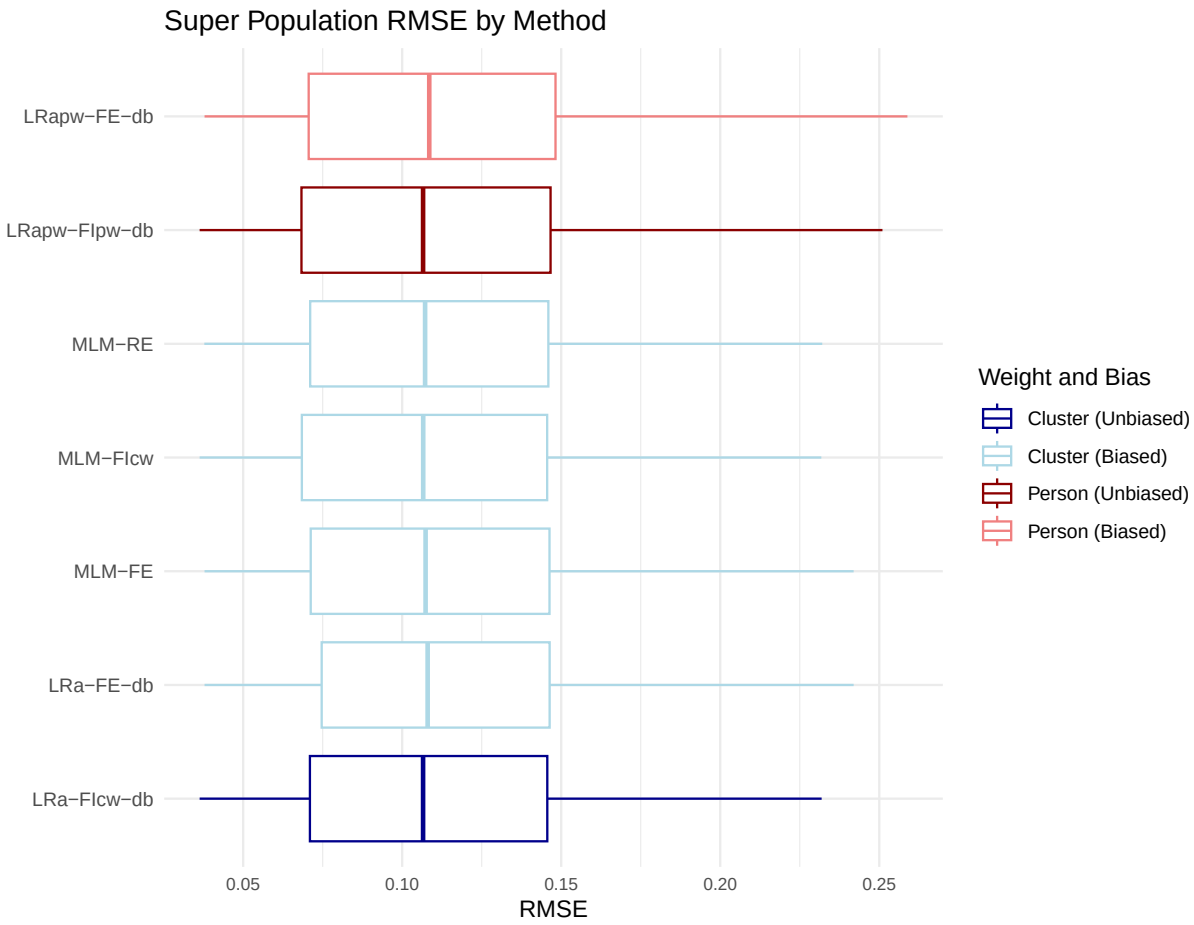
met hod	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FE-db	-0.134	0	0	0	L	L	0	L
LRapw-FE-db	-0.126	0	0	0	L	L	0	L
LRapw-FE-db	-0.114	0	L	0	0	0	0	L
MLM-FE	-0.148	0	0	0	L	L	0	L
MLM-FE	-0.143	S	0	L	S	S	0	L
MLM-FE	-0.14	0	L	0	S	L	L	L
MLM-FE	-0.126	0	0	0	L	L	0	L
MLM-FE	-0.122	0	L	L	L	L	0	L
MLM-RE	-0.132	0	0	0	L	L	0	L
MLM-RE	-0.127	S	0	L	S	S	0	L
MLM-RE	-0.117	0	L	0	S	L	L	L
MLM-RE	-0.113	0	0	0	L	L	0	L
MLM-RE	-0.11	S	L	0	L	0	0	L

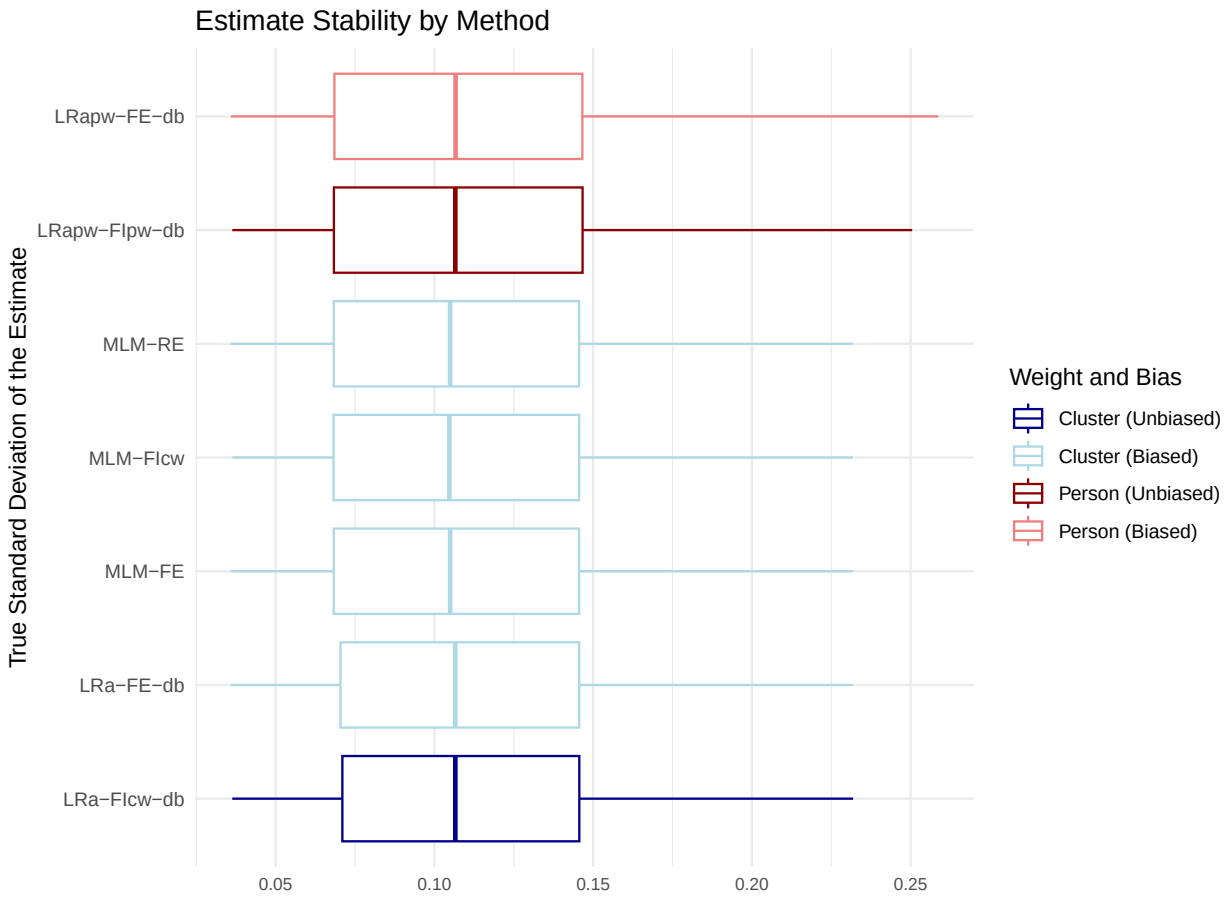
What does the RMSE look like for each estimator

As far as I can tell the RMSE actually looks pretty similar across estimators, which is consistent with the idea of there being a bias variance trade off. Unlike with the bias, the finite population RMSEs are fairly different from the super population RMSEs.

Like above, the boxplots are across the different data generating processes, each dot is a mean across simulation replications. The methods in light blue are those classified to be biased by theory. The ratio of finite RMSE to super RMSE plots show the ratio of RMSE from the finite population models over the super population models, and are designed to show that these are different.

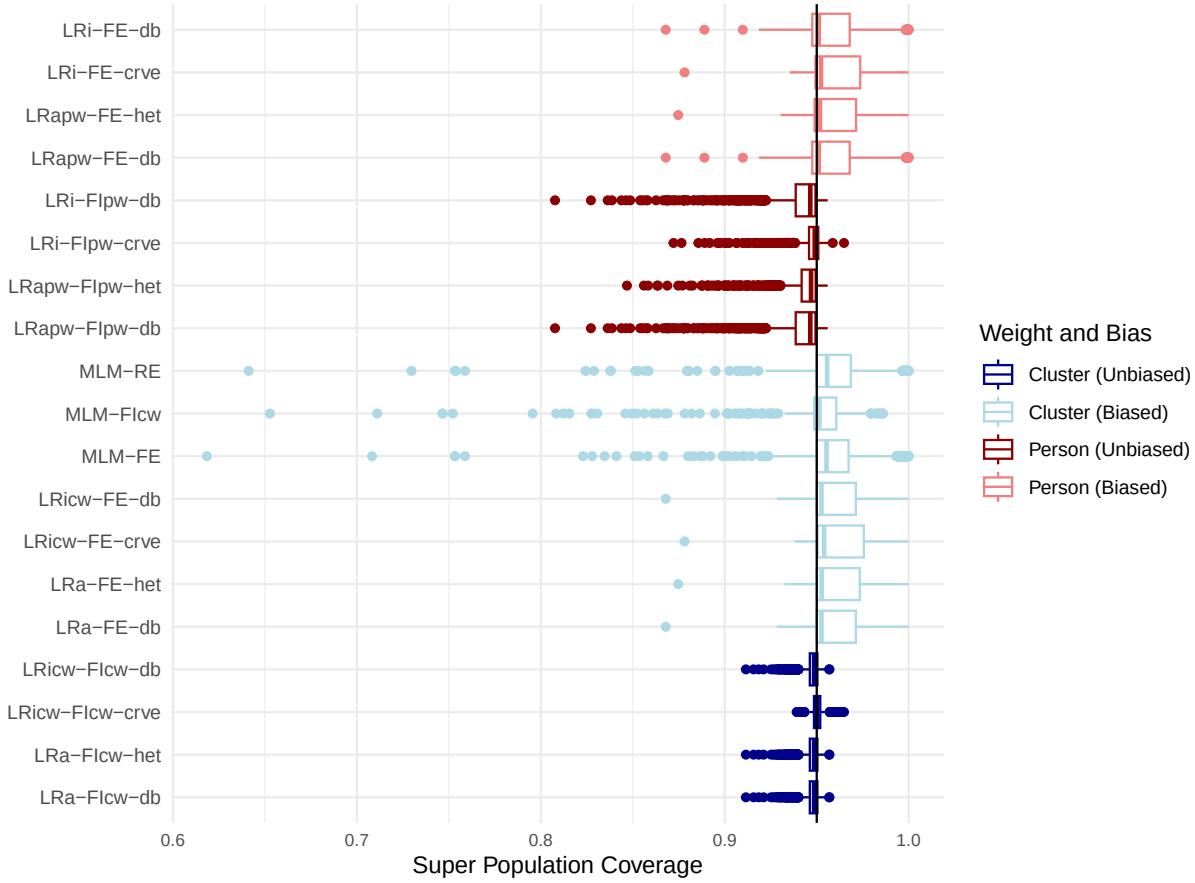
To-Do: Fix the graph to only include the core methods Re-Run the models here

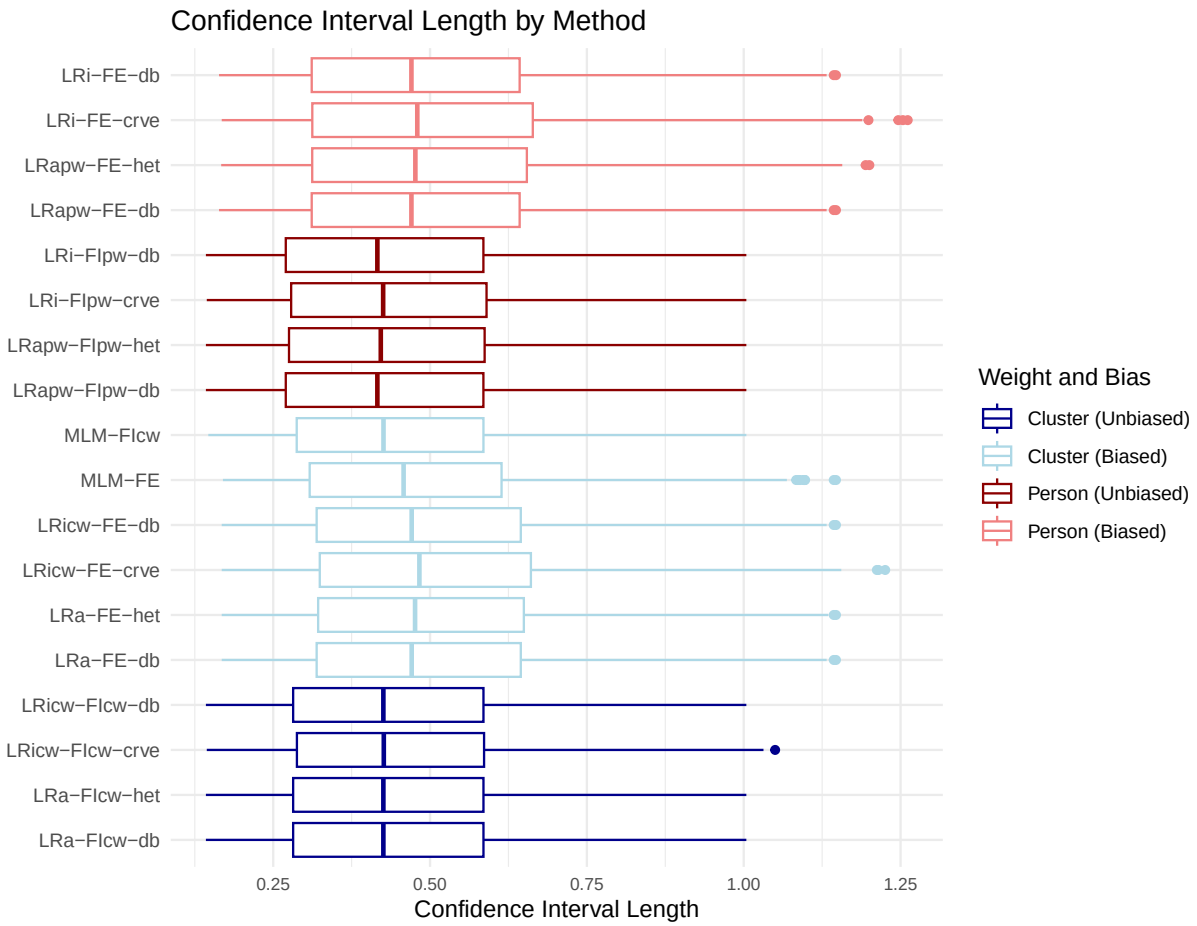

Super Population RMSE by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
0.05
0.10
0.15
0.20
0.25
RMSE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)



What does the coverage look like for each estimator

Coverage is the percentage of simulation replications where the estimator is within the 95% confidence interval. We would like the number to 95% (the boxplots have a red line at there). The notable things are that the coverage for the MLM estimators is often very bad but other than that the coverage for the cluster weighted estimators is generally good. The coverage rates for the person weighted estimators however are pretty bad across methods,it's either very low (meaning the SEs are being underestimated) or too high (meaning the SEs are being overestimated within the same methods. In addition, finite versus super makes very little difference for the cluster targeted estimators but makes a noticeable difference for the person weighted estimators.

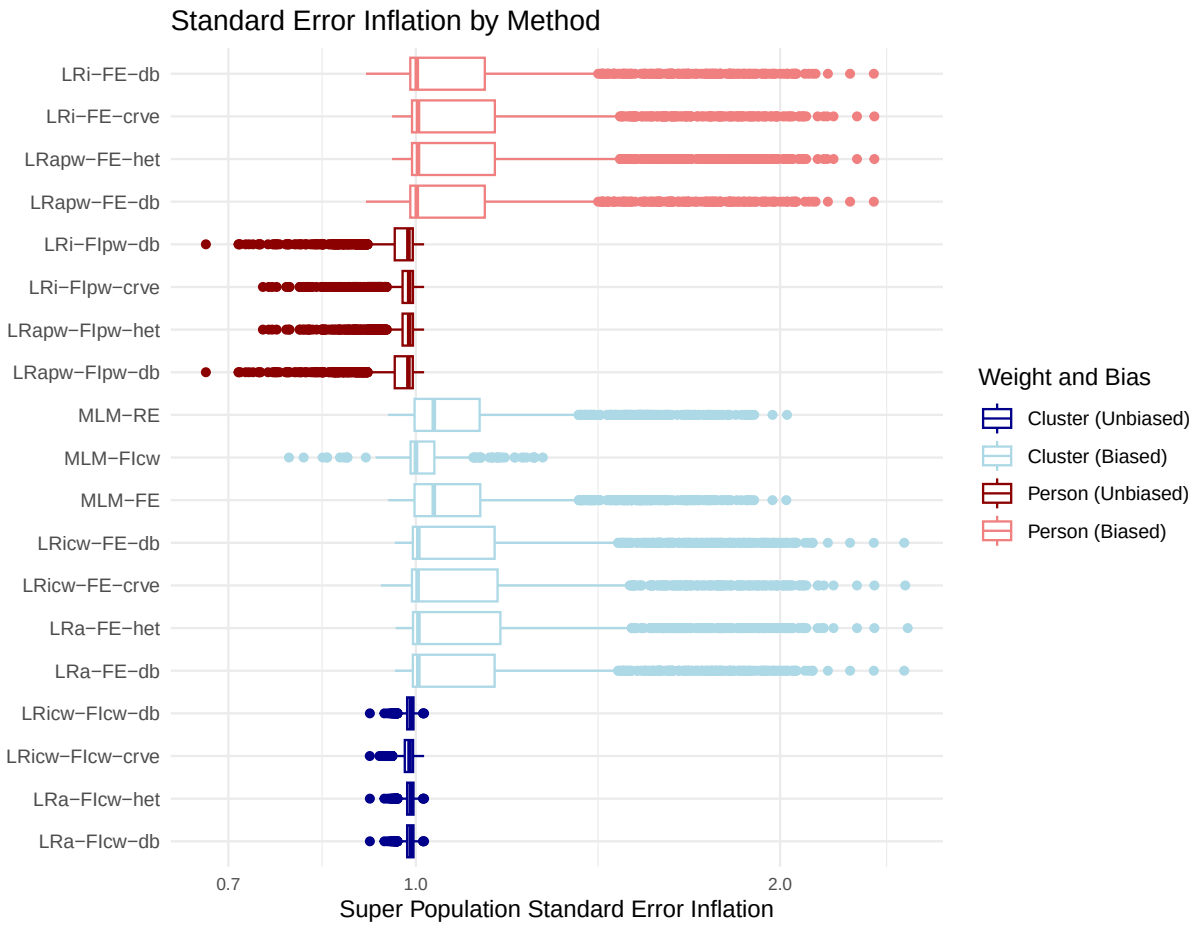

LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.6
0.7
0.8
0.9
1.0
Super Population Coverage
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

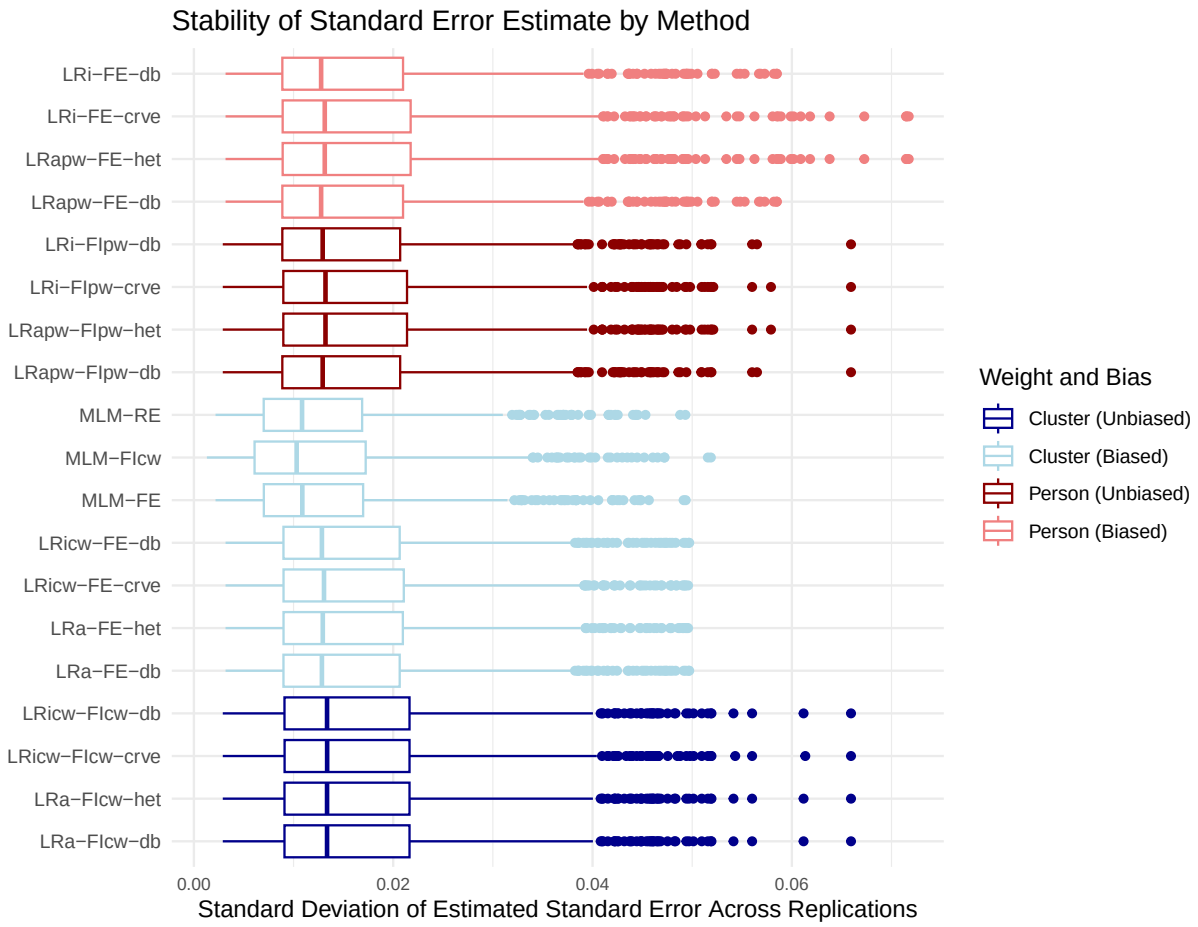


What do the standard errors look like for each estimator

Standard error inflation is the ration of the estimated standard error over the true standard error. High inflation means the standard errors are too big and low inflation means the standard errors are too small.

For some of the methods the standard error estimates are way off, but it does not seem obviously driven by low degrees of freedom. The true finite standard errors do not look to be the same as the true super standard errors.


Standard Error Inflation by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
1.0
2.0
Super Population Standard Error Inflation
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)



========== Model for method: LRa-FE-db Number of observations: 1296 R-squared: 0.7457939 ==========

========== Model for method: LRa-FE-het Number of observations: 1296 R-squared: 0.7486725 ==========

========== Model for method: LRa-FIcw-db Number of observations: 1296 R-squared: 0.728729 ==========

========== Model for method: LRapw-FE-db Number of observations: 1296 R-squared: 0.769057 ==========

========== Model for method: LRapw-FE-het Number of observations: 1296 R-squared: 0.7735951 ==========

========== Model for method: LRapw-FIpw-db Number of observations: 1296 R-squared: 0.7566025 ==========

========== Model for method: LRapw-FIpw-het Number of observations: 1296 R-squared: 0.7626567 ==========

========== Model for method: LRicw-FE-crve Number of observations: 1296 R-squared: 0.7464338 ==========

========== Model for method: LRicw-FIcw-crve Number of observations: 1296 R-squared: 0.7279335 ==========

========== Model for method: MLM-FE Number of observations: 1296 R-squared: 0.7635452 ==========

========== Model for method: MLM-FIcw Number of observations: 1296 R-squared: 0.7863796 ==========

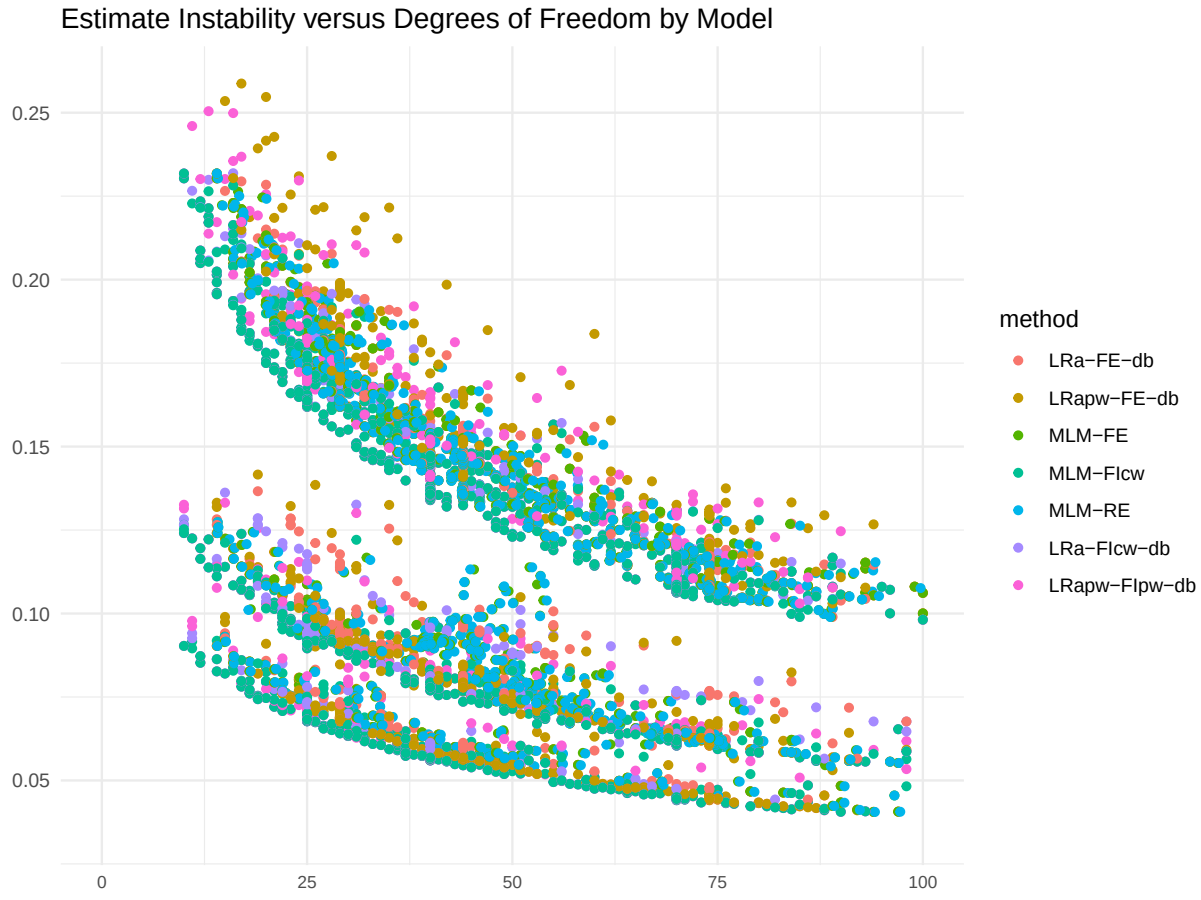
========== Model for method: MLM-RE Number of observations: 1296 R-squared: 0.7647602 ==========

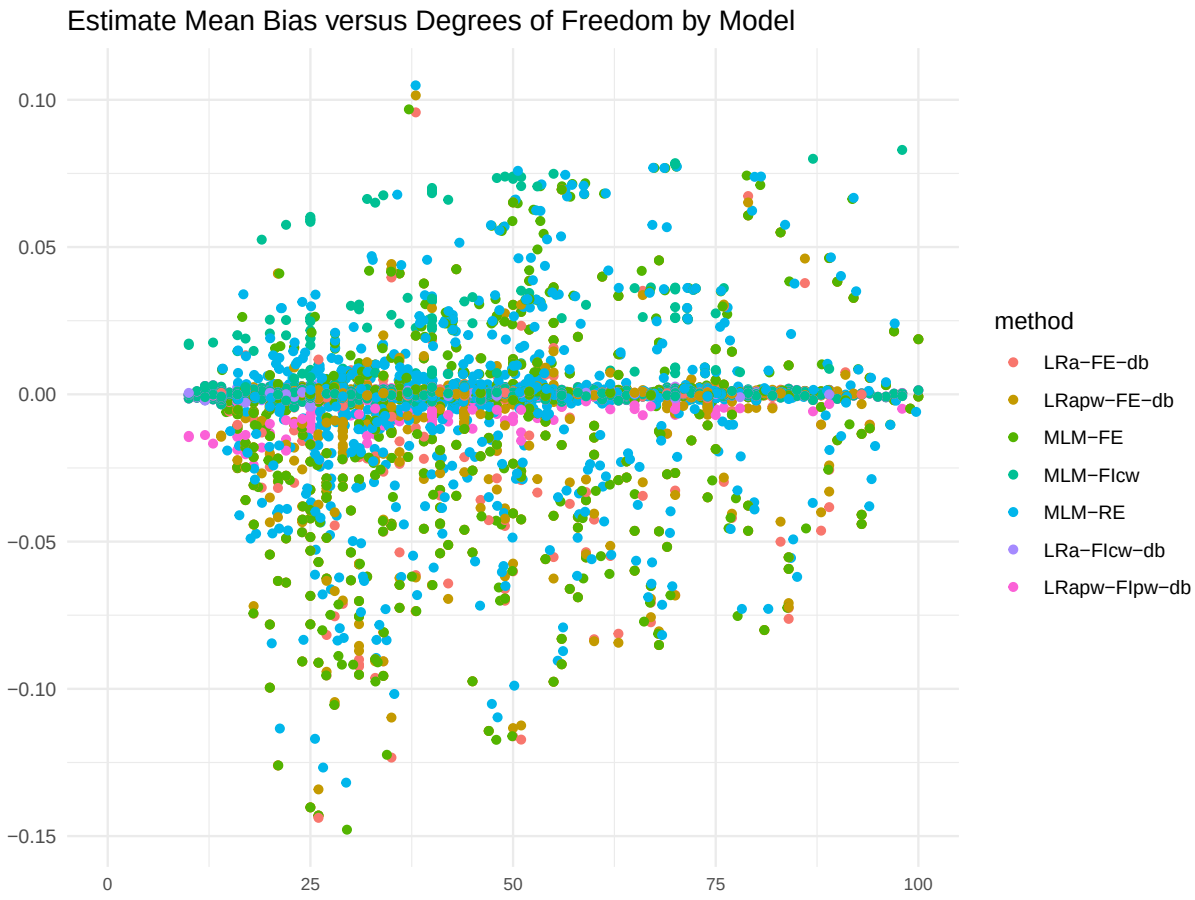
term LRa-FE-db LRa-FE-het LRa-FIcw-db

1 (Intercept) 0.01*** 0.01*** 0.01 ***2 block_typeS 0.00 0.00 0.00 3 block_typeL 0.01*** 0.01*** 0.00 **4 J.barS -0.00** -0.00** -0.00 **5 J.barL -0.01*** -0.01*** -0.01 ***6 size.impact.cor.schoolL -0.00 -0.00 -0.00 7 sd.schoolL 0.00*** 0.00*** 0.00 ***8 ICC.schoolL 0.01*** 0.01*** 0.01 ***9 variable.nTRUE 0.00 0.00* 0.00. 10 cv.schoolS 0.00 0.00 0.00 11 cv.schoolL 0.00* 0.00. 0.00. 12 num.cov.schoolS -0.00 -0.00 -0.00 13 num.cov.schoolL -0.00 -0.00 0.00 14 block_typeS:J.barS -0.00 -0.00 -0.00 15 block_typeL:J.barS -0.00 -0.00 0.00 16 block_typeS:J.barL -0.00 -0.00 -0.00 17 block_typeL:J.barL -0.00*** -0.00*** -0.00 *18 block_typeS:size.impact.cor.schoolL -0.00 -0.00 -0.00 19 block_typeL:size.impact.cor.schoolL 0.00 0.00 0.00 20 block_typeS:sd.schoolL 0.00 0.00 0.00 21 block_typeL:sd.schoolL 0.00 0.00 0.00 22 block_typeS:ICC.schoolL -0.00 -0.00 -0.00 23 block_typeL:ICC.schoolL 0.00*** 0.00*** 0.01 ***24 block_typeS:variable.nTRUE 0.00 0.00 0.00 25 block_typeL:variable.nTRUE 0.00 0.00 0.00 26 block_typeS:cv.schoolS -0.00 -0.00 -0.00 27 block_typeL:cv.schoolS 0.00 0.00 0.00 28 block_typeS:cv.schoolL -0.00 -0.00 -0.00 29 block_typeL:cv.schoolL 0.00 0.00 0.00*** *30 block_typeS:num.cov.schoolS 0.00 0.00 0.00 31 block_typeL:num.cov.schoolS -0.00 -0.00 -0.00 32 block_typeS:num.cov.schoolL 0.00 0.00 0.00 33 block_typeL:num.cov.schoolL -0.00 -0.00 -0.00 34 J.barS:size.impact.cor.schoolL -0.00 -0.00 -0.00 35 J.barL:size.impact.cor.schoolL -0.00 -0.00 -0.00 36 J.barS:sd.schoolL -0.00 -0.00 -0.00 37 J.barL:sd.schoolL -0.00 -0.00 -0.00 38 J.barS:ICC.schoolL -0.00*** -0.00*** -0.00 ***39 J.barL:ICC.schoolL -0.01*** -0.01*** -0.01 ***40 J.barS:variable.nTRUE -0.00 -0.00 -0.00 41 J.barL:variable.nTRUE -0.00 -0.00* -0.00 42 J.barS:cv.schoolS 0.00 0.00 0.00 43 J.barL:cv.schoolS -0.00 -0.00 -0.00 44 J.barS:cv.schoolL -0.00. -0.00. -0.00. 45 J.barL:cv.schoolL -0.00 -0.00* -0.00** 46 J.barS:num.cov.schoolS 0.00 0.00 0.00 47 J.barL:num.cov.schoolS 0.00 0.00 0.00 48 J.barS:num.cov.schoolL -0.00 -0.00 -0.00 49 J.barL:num.cov.schoolL -0.00 -0.00 -0.00 50 size.impact.cor.schoolL:sd.schoolL -0.00 -0.00 0.00 51 size.impact.cor.schoolL:ICC.schoolL 0.00* 0.00* 0.00. 52 size.impact.cor.schoolL:variable.nTRUE 0.00* 0.00* 0.00 *53 size.impact.cor.schoolL:cv.schoolS 0.00 0.00 -0.00 54 size.impact.cor.schoolL:cv.schoolL 0.00 0.00 0.00 55 size.impact.cor.schoolL:num.cov.schoolS 0.00 0.00 0.00 56 size.impact.cor.schoolL:num.cov.schoolL 0.00 0.00 0.00 57 sd.schoolL:ICC.schoolL -0.00 **-0.00** -0.00* 58 sd.schoolL:variable.nTRUE -0.00* -0.00* -0.00 *59 sd.schoolL:cv.schoolS -0.00 **-0.00** -0.00* 60 sd.schoolL:cv.schoolL -0.00* -0.00* -0.00 *61 sd.schoolL:num.cov.schoolS 0.00 0.00 -0.00 62 sd.schoolL:num.cov.schoolL -0.00 -0.00 0.00 63 ICC.schoolL:variable.nTRUE -0.00* -0.00* -0.00 *64 ICC.schoolL:cv.schoolS -0.00 -0.00 0.00 65 ICC.schoolL:cv.schoolL -0.00*** -0.00*** -0.00 ***66 ICC.schoolL:num.cov.schoolS -0.00 -0.00 -0.00 67 ICC.schoolL:num.cov.schoolL -0.00 -0.00 -0.00 68 variable.nTRUE:cv.schoolS -0.00 -0.00 -0.00 69 variable.nTRUE:cv.schoolL 0.00*** 0.00*** 0.01 ***70 variable.nTRUE:num.cov.schoolS -0.00 -0.00* -0.00 71 variable.nTRUE:num.cov.schoolL -0.00 -0.00 -0.00 72 cv.schoolS:num.cov.schoolS 0.00. 0.00. 0.00 73 cv.schoolL:num.cov.schoolS 0.00* 0.00* 0.00 74 cv.schoolS:num.cov.schoolL 0.00 0.00 0.00 75 cv.schoolL:num.cov.schoolL 0.00 0.00 0.00 LRapw-FE-db LRapw-FE-het LRapw-FIpw-db LRapw-FIpw-het LRicw-FE-crve 1 0.01*** 0.01*** 0.01*** 0.01 ***2 0.00 0.00 0.00 0.00 0.00 3 0.00*** 0.00*** 0.00** 0.01 ***4 -0.00*** -0.00. -0.00** -0.00 **5 -0.00** -0.00** -0.01*** -0.01*** -0.01 ***6 -0.00 -0.00 -0.00 -0.00 -0.00 7 0.00*** 0.00*** 0.00*** 0.00 ***8 0.01*** 0.01*** 0.01*** 0.01 ***9 -0.00 -0.00 -0.00 -0.00 0.00 10 -0.00 -0.00 -0.00 -0.00 0.00 11 -0.00. -0.00. -0.00 -0.00 0.00 ***12 -0.00 -0.00 -0.00 -0.00 -0.00 13 0.00 0.00 0.00 0.00 -0.00 14 -0.00 -0.00 -0.00 -0.00 -0.00 15 -0.00 -0.00 0.00 0.00 -0.00 16 -0.00 -0.00 -0.00 -0.00 -0.00 17 -0.00*** -0.00*** -0.00* -0.00. -0.00 ***18 -0.00 -0.00 -0.00 -0.00 -0.00 19 0.00 0.00 0.00 0.00 0.00 20 0.00 0.00 0.00 0.00 0.00 21 0.00 0.00 0.00 0.00 0.00 22 -0.00 -0.00 -0.00 -0.00 -0.00 23 0.00*** 0.00*** 0.00*** 0.00 ***24 0.00 0.00 0.00 0.00 0.00 25 0.00*** 0.00*** -0.00 -0.00 0.00 26 0.00 0.00 -0.00 -0.00 -0.00 27 0.00. 0.00. 0.00 0.00 0.00 28 0.00 0.00 -0.00 -0.00 -0.00 29 0.00*** 0.00*** 0.00 0.00 0.00 30 0.00 0.00 0.00 0.00 0.00 31 -0.00 -0.00 -0.00 -0.00 -0.00 32 0.00 0.00 0.00 0.00 0.00 33 -0.00

-0.00 -0.00 -0.00 -0.00 34 -0.00 -0.00 -0.00 -0.00 -0.00 35 -0.00 -0.00 -0.00 -0.00 -0.00 36 -0.00 -0.00 -0.00 -0.00 -0.00 37 -0.00 -0.00 -0.00 -0.00 -0.00 38 -0.00*** -0.00*** -0.00*** -0.00*** -0.00 ***39 -0.01*** -0.01*** -0.01*** -0.01*** -0.01 ***40 -0.00 -0.00 0.00 0.00 -0.00 41 -0.00 -0.00*** -0.00 -0.00. -0.00 *42 0.00 -0.00 0.00 0.00 0.00 0.00 43 -0.00 -0.00 0.00 -0.00 -0.00 44 -0.00 -0.00* -0.00 -0.00 -0.00. 45 -0.00* -0.00*** -0.00 -0.00. -0.00 *46 0.00 0.00 0.00 0.00 0.00 47 0.00 0.00 0.00 0.00 0.00 48 -0.00 -0.00 -0.00 -0.00 -0.00 49 -0.00 -0.00 -0.00 -0.00 -0.00 50 -0.00 -0.00 -0.00 -0.00 -0.00 51 0.00* 0.00. 0.00. 0.00 *52 0.00 0.00. 0.00. 0.00. 0.00* 53 0.00 0.00 -0.00 -0.00 0.00 54 0.00 0.00 0.00 0.00 0.00 55 0.00 0.00 0.00 0.00 0.00 56 0.00 0.00 0.00 0.00 0.00 57 -0.00*** -0.00** -0.00*** -0.00*** -0.00 **58 -0.00 -0.00 -0.00 -0.00 -0.00 *59 -0.00* -0.00* -0.00* -0.00* -0.00** 60 -0.00 -0.00 -0.00 -0.00 -0.00 *61 0.00 0.00 -0.00 -0.00 0.00 62 -0.00 -0.00 0.00 -0.00 -0.00 63 0.00*** 0.00*** 0.00*** 0.00*** -0.00 **64 0.00 0.00 0.00 0.00 -0.00 65 0.00*** 0.00*** 0.00* 0.00** -0.00 ***66 -0.00 -0.00 -0.00 -0.00 -0.00 67 -0.00 -0.00 -0.00 -0.00 -0.00 68 0.00* **0.00** 0.00 0.00 -0.00 69 0.01*** 0.01*** 0.00*** 0.00*** 0.00 ***70 -0.00* **-0.00* -0.00* -0.00* -0.00** ***71 -0.00 -0.00 -0.00 -0.00 -0.00 72 0.00 0.00 0.00* **0.00* 0.00. 73 0.00* 0.00* 0.00* 0.00* 0.00** ***74 0.00 0.00 0.00 0.00 0.00 75 0.00 0.00 0.00 0.00 0.00 LRicw-FIcw-crve MLM-FE MLM-FIcw MLM-RE 1 0.01*** 0.01*** 0.01 ***2 0.00 0.00 0.00 0.00 3 0.00*** *0.01*** 0.00 0.01 ***4 -0.00*** *-0.00* -0.00* -0.00 *5 -0.01*** -0.00*** -0.00*** -0.00 ***6 -0.00 -0.00 -0.00 -0.00 7 0.00*** 0.01*** 0.01*** 0.01 ***8 0.01*** 0.01*** 0.02*** 0.01 ***9 0.00* **0.00 0.00 0.00 10 0.00 0.00 0.00 0.00 11 0.00* 0.00 0.00 0.00 12 -0.00 -0.00 -0.00 -0.00 13 0.00 -0.00 -0.00 -0.00 14 -0.00 -0.00 -0.00 -0.00 15 0.00 -0.00 0.00 -0.00 16 -0.00 -0.00 -0.00 -0.00 17 -0.00. -0.00*** -0.00 -0.00 ***18 -0.00 -0.00 -0.00 -0.00 19 0.00 0.00 0.00 0.00 20 0.00 -0.00 -0.00 -0.00 21 0.00 -0.00. 0.00 -0.00. 22 0.00 -0.00 0.00 -0.00 23 0.01*** 0.00 0.00*** 0.00 24 0.00 0.00 0.00 0.00 25 0.00 0.00 0.00 0.00 26 -0.00 -0.00 -0.00 -0.00 -0.00 27 0.00 0.00 0.00 0.00 28 -0.00 -0.00 -0.00 -0.00 29 0.00* 0.00 0.00 0.00 30 0.00 0.00 0.00 0.00 31 -0.00 -0.00 -0.00 -0.00 32 0.00 0.00 0.00 0.00 33 -0.00 -0.00 -0.00 -0.00 34 -0.00 -0.00 -0.00 -0.00 35 -0.00 -0.00. -0.00* -0.00. 36 -0.00 -0.00* -0.00** -0.00 *37 -0.00 -0.00*** -0.00*** -0.00 ***38 -0.00*** -0.00*** -0.01*** -0.00 ***39 -0.01*** -0.01*** -0.01 ***40 -0.00 -0.00 -0.00 -0.00 41 -0.00* **-0.00 -0.00. -0.00 42 0.00 0.00 0.00 0.00 43 -0.00 0.00 0.00 0.00 44 -0.00. -0.00 -0.00 -0.00 45 -0.00** -0.00 -0.00 -0.00 46 0.00 0.00 0.00 0.00 47 0.00 0.00 0.00 0.00 48 -0.00 0.00 0.00 0.00 49 -0.00 -0.00 -0.00 -0.00 50 0.00 -0.00 -0.00 -0.00 51 0.00. 0.00. 0.00. 0.00. 52 0.00* 0.00* 0.00* 0.00 *53 -0.00 0.00 0.00 0.00 54 0.00 0.00 0.00 0.00 0.00 55 0.00 0.00 0.00 0.00 56 0.00 0.00 0.00 0.00 57 -0.00* -0.00*** -0.00*** -0.00 ***58 -0.00* **-0.00. -0.00 -0.00 59 -0.00* -0.00* -0.00. -0.00. 60 -0.00* -0.00 -0.00 -0.00 61 -0.00 0.00 0.00 0.00 62 0.00 -0.00 -0.00 -0.00 63 -0.00** 0.00 0.00 0.00 64 -0.00 -0.00 -0.00 -0.00 65 -0.00*** -0.00 -0.00 -0.00 66 -0.00 -0.00 -0.00 -0.00 67 -0.00 -0.00 -0.00 -0.00 68 0.00 -0.00 0.00 -0.00 69 0.01*** 0.00 0.00* 0.00 70 -0.00* -0.00* -0.00. -0.00 *71 -0.00 -0.00 -0.00 -0.00 72 0.00* 0.00 0.00 0.00 73 0.00* 0.00* 0.00* 0.00* 74 0.00 0.00 -0.00 0.00 75 0.00 0.00 0.00 0.00 method sd.SE.hat.s sd.ATE size.impact.cor J.bar sd.school 1 LRa-FE-db 0.04969950 L L 0 L 2 LRa-FE-db 0.04964257 L L 0 0 3 LRa-FE-db 0.04934965 L L 0 L 4 LRa-FE-db 0.04914330 L L 0 L 5 LRa-FE-db 0.04833746 L L S 0 6 LRa-FE-het 0.04958679 L L 0 L 7 LRa-FE-het 0.04945717 L L 0 0 8 LRa-FE-het 0.04934965 L L 0 L 9 LRa-FE-het 0.04914330 L L 0 L 10 LRa-FE-het 0.04903156 L L S 0 11 LRapw-FE-db 0.05849423 L L 0 L 12 LRapw-FE-db 0.05847556 L L 0 L 13 LRapw-FE-db 0.05820059 L L 0 0 14 LRapw-FE-db 0.05727228 L L 0 0 15 LRapw-FE-db 0.05686273 L L 0 0 16 LRapw-FE-het 0.07172370 L L 0 L 17 LRapw-FE-het 0.07150984 L L 0 L 18 LRapw-FE-het 0.07145919 L L 0 0 19 LRapw-FE-het 0.06726747 L L 0 0 20 LRapw-FE-het 0.06379074 L L 0 L 21 LRicw-FE-crve 0.04958679 L L 0 L 22 LRicw-FE-crve 0.04934965 L L 0 L 23 LRicw-FE-crve 0.04930498 L L 0 0 24 LRicw-FE-crve 0.04914330 L L 0 L 25 LRicw-FE-crve 0.04903156 L L S 0 26 MLM-FE 0.04934962 L L 0 L 27 MLM-FE 0.04914328 L L 0 L 28 MLM-FE 0.04565466 L L 0 0 29 MLM-FE 0.04485496 L L 0 L 30 MLM-FE 0.04478190 L L 0 L 31 MLM-FIcw 0.05187094 L L 0 L 32 MLM-FIcw 0.05157175 L L 0 L 33 MLM-FIcw 0.04722937 L L 0 L 34 MLM-FIcw 0.04718535 L L 0 0 35 MLM-FIcw 0.04653229 L L 0 L 36 MLM-RE 0.04931761 L L 0 L 37 MLM-RE 0.04879479 L L 0 L 38 MLM-RE 0.04531317 L L 0 0 39 MLM-RE 0.04454339 L L 0 L 40 MLM-RE 0.04436160 L L 0 L 41 LRa-FIcw-db 0.06593430 L L S 0 42 LRa-FIcw-db 0.06117638 L L 0 L 43 LRa-FIcw-db 0.05601163 L L 0 L 44 LRa-FIcw-db 0.05414307 L L 0 L 45 LRa-FIcw-db 0.05193417 L L 0 L 46 LRapw-FIpw-db 0.06593430 L L S 0 47 LRapw-FIpw-db 0.05650179 L L 0 L 48 LRapw-FIpw-db 0.05601163 L L 0 L 49 LRapw-FIpw-db 0.05193417 L L 0 L 50 LRapw-FIpw-db 0.05187420 L L 0 L 51 LRapw-FIpw-het 0.06593430 L L S 0 52 LRapw-FIpw-het 0.05790380 L L 0 L 53 LRapw-FIpw-het 0.05601163 L L 0 L 54 LRapw-FIpw-het 0.05211906 L L 0 L 55 LRapw-FIpw-het 0.05193417 L L 0 L 56 LRicw-FIcw-crve 0.06593430 L L S 0 57 LRicw-FIcw-crve 0.06134810 L L 0 L

58 LRicw-FIcw-crve 0.05601163 L L 0 L 59 LRicw-FIcw-crve 0.05431772 L L 0 L 60 LRicw-FIcw-crve 0.05193417 L L 0 L ICC.school cv.school num.cov.school proptx.impact.cor size.intercept.cor 1 L S 0 L L 2 L S L L L 3 L 0 L L L 4 L 0 0 L L 5 L S S L L 6 L S 0 L L 7 L S L L L 8 L 0 L L L 9 L 0 0 L L 10 L S S L L 11 L L 0 L L 12 L L L L L 13 L L L L L 14 L L 0 L L 15 L S L L L 16 L L 0 L L 17 L L L L L 18 L L L L L 19 L L 0 L L 20 L L 0 L L 21 L S 0 L L 22 L 0 L L L 23 L S L L L 24 L 0 0 L L 25 L S S L L 26 L 0 L L L 27 L 0 0 L L 28 L L L L L 29 L S 0 L L 30 L L 0 L L 31 L 0 L L L 32 L 0 0 L L 33 L L 0 L L 34 L L L L L 35 L S 0 L L 36 L 0 L L L 37 L 0 0 L L 38 L L L L L 39 L L 0 L L 40 L S 0 L L 41 L S S L L 42 L L L L L 43 L S S L L 44 L L 0 L L 45 L S 0 L L 46 L S S L L 47 L L L L L 48 L S S L L 49 L S 0 L L 50 L 0 L L L 51 L S S L L 52 L L L L L 53 L S S L L 54 L L 0 L L 55 L S 0 L L 56 L S S L L 57 L L L L L 58 L S S L L 59 L L 0 L L 60 L S 0 L L size.impact.cor.school block_type 1 0 L 2 L L 3 L L 4 L L 5 0 L 6 0 L 7 L L 8 L L 9 L L 10 0 L 11 L L 12 L L 13 L L 14 L L 15 L L 16 L L 17 L L 18 L L 19 L L 20 0 L 21 0 L 22 L L 23 L L 24 L L 25 0 L 26 L L 27 L L 28 L L 29 0 L 30 L L 31 L L 32 L L 33 L L 34 L L 35 L L 36 L L 37 L L 38 L L 39 L L 40 0 L 41 0 L 42 L L 43 0 L 44 L L 45 0 L 46 0 L 47 L L 48 0 L 49 0 L 50 L L 51 0 L 52 L L 53 0 L 54 L L 55 0 L 56 0 L 57 L L 58 0 L 59 L L 60 0 L




Estimate Mean Bias versus Degrees of Freedom by Model
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
25
50
75
100
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db